SCOPE 3 — UX, DESIGN SYSTEM, FRONTEND DELIVERY, AND EXPERIENCE STANDARDS
(Rosotravel — Premium Decision-First Travel Brand)

3.0 Purpose, Positioning, and Scope Boundaries

Scope 3 defines the global UX, visual system, interaction logic, and frontend delivery standards for the new Rosotravel website.
Its objective is to deliver:
- a best-in-class tours & activities purchase experience,
- a mobile-first high-performance frontend,
- a modular design system that supports programmatic landing pages,
- and a premium, decision-first brand expression that clearly differentiates Rosotravel from generic marketplaces.

Rosotravel must not be designed as:
- a generic catalog of tours,
- a crowded OTA marketplace,
- or an ultra-luxury concierge product for a very narrow niche.

Rosotravel should be designed as:
- a premium global travel expert,
- a curated decision platform,
- and a highly usable booking product that helps travelers choose with confidence.

This means the UX must communicate immediately that:
1. Rosotravel already did the difficult comparison work.
2. Rosotravel takes responsibility for what it recommends.
3. Rosotravel is premium, but still easy and familiar to use.
4. Rosotravel helps the user move from uncertainty to decision without overwhelm.

The target user should feel within seconds:
- this is higher quality than a standard marketplace,
- the site is easy to understand,
- the shortlist exists for a reason,
- and booking here feels safer, more considered, and more premium.

Scope 3 must remain fully compatible with:
Scope 1:
- Viator API integration and required booking/content elements on specific landing pages.

Scope 2:
- Programmatic SEO requirements,
- page types and template logic,
- indexation, canonical, hreflang, page eligibility,
- modular content governance,
- and all decision-platform content modules already defined in the specification.

Important:
Scope 3 does not replace Scope 1 or Scope 2.
It translates them into:
- customer journey UX,
- interaction rules,
- visual hierarchy,
- and frontend system behavior.

3.1 Architecture Constraints and Responsibilities

3.1.1 Backend and Admin
The existing backend and admin panel remain in Laravel.
The vendor must not introduce WordPress or any WordPress-based CMS as part of this project.

The existing backend/admin environment remains the operational source of truth for:
- product data,
- destination data,
- SEO state,
- module configuration,
- content lock states,
- workflow statuses,
- and system-level governance.

3.1.2 Frontend
The public website frontend will be rebuilt from scratch with new code.
The frontend must support HTML-first rendering and a production approach compatible with:
- SSR,
- SSG,
- and/or ISR,
depending on the final stack and template needs.

The frontend must be:
- mobile-first,
- SEO-safe,
- performance-optimized,
- and compatible with programmatic publishing at scale.

[PATCH] — Frontend Technology Direction, Mobile-First Priority, Migration Equity, and Build Transparency

INSERT under section 3.1 Architecture Constraints and Responsibilities
Recommended placement:
- after 3.1.2 Frontend
- before 3.1.3 Source of Truth for SEO Rendering

3.1.2A Frontend Technology Direction

Rosotravel agrees that a modern React-based frontend framework with strong SEO and rendering capabilities is the correct direction for this project.

Preferred direction:
- Next.js + TypeScript

Reason:
- strong support for SSR / SSG / ISR,
- good fit for programmatic SEO,
- strong ecosystem maturity,
- high-performance rendering possibilities,
- and compatibility with modular frontend architecture.

Important:
Next.js + TypeScript is the preferred direction, but Rosotravel remains open to another equivalent frontend architecture if the vendor can clearly prove that the proposed alternative provides:
- equal or better mobile-first performance,
- equal or better SEO rendering control,
- equal or better compatibility with programmatic publishing,
- equal or better integration safety with the Laravel backend,
- and equal or better long-term maintainability.

Any alternative stack must be justified explicitly by the vendor.

3.1.2B Mobile-First Requirement (Strategic)
The frontend must be designed and implemented mobile-first.

This is a hard product requirement because a large share of users first enter Rosotravel from mobile search traffic.
Therefore:
- mobile experience is not a responsive afterthought,
- mobile hierarchy must be primary,
- mobile rendering performance must be prioritized,
- and key destination and product pages must be designed first for mobile understanding and conversion.

Desktop optimization remains important, but mobile must lead the design and engineering approach.

3.1.2C Frontend Build Transparency — Custom vs Template / Starter Usage
The vendor must explicitly disclose how the frontend will be built.

This must include:
- whether the frontend is fully custom,
- whether any starter framework, template, boilerplate, or purchased theme is used,
- whether any external design system or component library is used,
- and which parts are custom-built vs reused.

Rosotravel's expectation is a premium, decision-first, fully branded frontend.
Therefore:
- the final user-facing experience must not look like a reused travel template,
- and any starter/base tooling may only be used as a technical accelerator, not as a visual or structural limitation.

The vendor must clearly state:
- what is custom,
- what is framework-standard,
- what is third-party,
- and how final ownership and maintainability of the frontend codebase will work.

INSERT under section 3.1.3 Source of Truth for SEO Rendering or directly after it

3.1.3A SEO Equity Preservation During Migration
A robust SEO migration plan is mandatory.

The new frontend must preserve as much existing SEO equity as possible during migration from the current website to the new architecture.

At minimum, the migration approach must include:
- a full 301 redirect mapping strategy from current URLs to the new URL architecture,
- preservation or controlled migration of metadata,
- preservation or controlled migration of structured data,
- preservation or controlled migration of canonical logic,
- preservation or controlled migration of hreflang where applicable,
- and controlled rollout / QA for key organic landing pages.

The frontend and migration plan must be implemented in a way that minimizes:
- traffic loss,
- ranking volatility,
- broken deep links,
- and structured-data regressions.

INSERT under section 3.1 Architecture Constraints and Responsibilities, after SEO rendering or before 3.2 Customer Journey

3.1.4 Integration Risk and API Layer Requirement
The primary technical risk in this frontend rebuild is the interface between the new frontend and the existing Laravel backend.

The vendor must assume that this integration layer is critical and must be treated as a first-class architecture topic.

The solution must ensure:
- a seamless API contract between frontend and backend,
- deterministic rendering inputs for SEO and page composition,

- stable content delivery for programmatic pages,
- safe handling of module configuration and page state,
- and a clear separation between frontend rendering responsibilities and backend source-of-truth responsibilities.

The vendor must describe:
- how the frontend will consume Laravel data,
- how SSR / SSG / ISR rendering will be powered,
- how cache invalidation will work,
- and how frontend-backend contract changes will be managed safely over time.

INSERT under section 3.4 Mobile vs Desktop Optimization and Experimentation
Recommended placement:
- after 3.4.1 Mobile and Desktop UX

3.4.1A Mobile Search Entry Reality
The vendor must design key decision pages with the assumption that many users first arrive from mobile search, not from desktop direct traffic.

Therefore, special attention is required for:
- first-screen clarity,
- speed of understanding,
- mobile trust placement,
- mobile CTA logic,
- mobile-friendly shortlist presentation,
- and low-friction continuation into booking.

This requirement applies especially to:
- City pages
- Things-to-do pages
- Attraction pages
- Product pages
- Bundle pages

END OF PATCH

3.1.3 Source of Truth for SEO Rendering
Canonical URLs, hreflang, index/noindex, structured data, and other SEO governance outputs must be rendered deterministically by the frontend based on backend-provided SEO state.

The frontend must not invent SEO state client-side.
All SEO-critical rendering decisions must be:
- deterministic,
- backend-controlled,
- and auditable.

3.1.4 Frontend responsibility boundaries
The frontend is responsible for:
- rendering page templates,
- rendering design system components,
- applying safe runtime presentation logic where defined,
- respecting SEO and content locks,
- and implementing the purchase and post-purchase UX.

The frontend is not allowed to:
- break canonical logic,
- bypass page eligibility rules,
- change SEO-critical content structure unpredictably,
- or create ungoverned dynamic URL variants.

3.2 Customer Journey and Conversion UX (Critical)

3.2.1 Purchase UX Standard
The vendor must design and implement a purchase experience aligned with best-in-class tours & activities booking standards.

The UX must maximize conversion across the full funnel:
- discovery,
- evaluation and comparison,
- selection,
- checkout,
- confirmation,
- post-purchase experience.

The purchase UX should feel:
- premium,
- familiar,
- fast,
- trustworthy,
- and simple to understand.

Rosotravel must not look like a discount marketplace or an overloaded search results engine.
The purchase flow must support a curated, expert-led brand while remaining recognizable to users accustomed to travel booking standards.

3.2.2 Decision-first UX requirement
Rosotravel is a decision platform.
Therefore the UX must not be designed around “showing everything”.
It must be designed around:
- helping the traveler choose,
- reducing duplicate noise,
- explaining why certain options are surfaced,
- surfacing trade-offs where relevant,

- and making the next step obvious.

This must be visible across:
- city pages,
- things-to-do pages,
- attraction pages,
- product pages,
- bundle pages,
- and near pages.

The frontend must support:
- curated shortlists,
- decision blocks,
- trust modules,
- comparison clarity,
- and premium reassurance.

3.2.3 Mandatory Landing Page Elements (Scope 1 and Scope 2)
Scope 1 and Scope 2 specify required elements that must appear on particular page types:
- City,
- Attraction / POI,
- Product,
- Things-to-do,
- Near Pages,
- Bundles / Packages,
- Blog / Guides,
- Experts,
- Explore,
- Country,
- Region.

These elements are mandatory.
However, the vendor may propose improved design patterns and placement to maximize conversion, provided:
- the elements remain present and accessible on the intended landing pages,
- the final UX remains consistent with best practices,
- and the final implementation does not break programmatic SEO or decision-platform logic.

3.2.4 Reading vs planning vs booking modes
The UX must respect different user mental states:
- browsing,
- comparing,
- planning,
- booking.

The site must not force booking too early.
It must support:
- exploration,

- shortlist understanding,
- save / return-later behaviors,
- and then booking.

This includes:
- bundles,
- compare-friendly layouts,
- trust modules,
- and optional “email to self” or planning hooks where appropriate.

3.2.5 Founder accountability and trust
Rosotravel’s UX should communicate responsibility.
Users should feel:
- this brand stands behind its recommendations,
- it does not simply list and disappear,
- and it takes responsibility for what it suggests.

A founder or brand-trust video may be used as a secondary trust layer, not as an interruptive sales device.
Its role is:
- accountability,
- credibility,
- and explanation of the Rosotravel standard.

Recommended placements:
- homepage trust section,
- About Rosotravel / About Us,
- selected brand-expertise sections,
- selected high-trust surfaces below core decision content.

Not recommended:
- autoplay hero on product pages,
- intrusive modal,
- or anything that interrupts booking.

3.3 Design System and Modularity Requirements

3.3.1 Modular Layout System
The frontend must be built as a modular system where page sections are implemented as reusable components and assembled per page type.

Components must support:
- consistent structure,
- predictable behavior,
- configurable variants,
- and scalable reuse across templates.

3.3.2 CMS and Marketing Flexibility

The system must allow the marketing team to:
- reorder or swap predefined modules within allowed constraints,
- update module variants for experiments,
- update trust and CTA modules,
- and introduce new modules in the future without rebuilding the entire frontend.

This flexibility must operate inside guardrails.
Marketing flexibility must not break:
- SEO governance,
- template identity,
- or content logic.

3.3.3 Consistency Across Page Types
The design system must support consistent templates and reusable components across:
- country,
- region,
- city,
- attraction / POI,
- product,
- bundle / package,
- blog / guides,
- experts,
- explore.

Consistency does not mean visual sameness.
It means:
- shared interaction logic,
- shared brand language,
- consistent component behavior,
- and a predictable decision journey.

3.3.4 Brand and visual direction
Rosotravel must feel premium, curated, calm, and expert-led.

It should not feel:
- cheap,
- crowded,
- promotional,
- noisy,
- over-designed,
- or luxury-theatrical.

Recommended design direction:
- premium Western digital quality,
- rounded corners rather than sharp square geometry,
- soft but controlled spacing,
- elegant restraint,
- high readability,

- strong hierarchy,
- premium but familiar booking patterns.

The interface should feel:
- higher quality than a standard marketplace,
- easier to trust,
- and more edited and intentional.

It should not feel like:
- a mass-market listing site,
- or a hyper-luxury bespoke travel concierge.

3.3.5 Geometry and component language
Recommended component language:
- rounded cards,
- rounded buttons,
- controlled shadows,
- clean image framing,
- generous white space,
- strong but quiet hierarchy,
- minimal visual clutter.

If a marketplace feels boxy and dense, Rosotravel should feel smoother, calmer, and more premium.

3.3.6 Decision hierarchy in design
Designers must visually distinguish:
A) decision blocks
- top picks,
- bundles,
- our picks,
- rails,
- trade-offs,
- “why these picks”

B) trust blocks
- ratings,
- recent proof,
- guide / team anchors,
- reviews,
- reputation strips

C) support blocks
- FAQ,
- inspiration,
- editorial,
- deeper browse flows

Primary actions must remain obvious.
Exploration must not compete visually with the main decision.

3.3.7 Premium without excess
Rosotravel is premium, not ultra-luxury.
Therefore:
- no heavy luxury clichés,
- no dark opulence as default,
- no over-formal concierge feel,
- no “status friction” design.

The brand should feel:
- elevated,
- accessible,
- expert,
- globally credible,
- and easy to use.

3.4 Mobile vs Desktop Optimization and Experimentation

3.4.1 Mobile and Desktop UX
The vendor must confirm and implement device-optimized patterns within responsive layouts to maximize conversion.
Mobile UX must be treated as primary.

This means:
- mobile-first hierarchy,
- mobile-first CTA strategy,
- mobile-safe component density,
- and strong tap-target usability.

Desktop may introduce:
- wider comparison surfaces,
- more visible trust sidebars,
- or richer contextual modules,
but must remain consistent with the same decision logic.

3.4.2 A/B Testing Capability (Mandatory)
The solution must support A/B testing and experimentation for marketing, including:
- alternative module order and placement,
- trust elements and conversion components,
- hero variants and above-the-fold layouts,
- microcopy and CTA variants,
- optional reassurance and hook modules.

The vendor must propose a practical experimentation approach that marketing can use without engineering work for every test, within defined guardrails.

Hard rule:
Testing flexibility must not allow:
- accidental SEO divergence,
- template chaos,
- or unsafe content variation outside approved slots.

3.4.3 Personalization guardrails
Where adaptive presentation is used, it must remain:
- subtle,
- deterministic,
- session-safe,
- and SEO-safe.

No UX variant may:
- break canonical layout identity,
- create visible mode-switching,
- or make the site feel unstable.

3.5 Media Requirements

3.5.1 Product Media
Product pages must support video in addition to photo galleries where video assets exist.

Media must be implemented with:
- mobile performance in mind,
- lazy loading,
- click-to-play patterns where appropriate,
- poster image fallbacks,
- and stable layout allocation.

3.5.2 Media quality direction
Media should feel:
- real,
- aspirational,
- warm,
- human,
- destination-rich,
- and premium.

It should not feel:
- cheap,
- stock-heavy,
- noisy,
- or social-first in a low-trust way.

3.5.3 Founder and brand media
Founder video and other brand-trust media should be treated as optional trust modules.
They must be:

- performance-safe,
- lazy loaded,
- subtitle-ready,
- and never block the primary decision flow.

3.5.4 Media and design consistency
Media treatment must be consistent with premium brand goals:
- clean framing,
- controlled overlays,
- no aggressive autoplay,
- no visual overload,
- and high-quality transitions.

3.6 AMP (Accelerated Mobile Pages)

3.6.1 AMP Decision Rule
AMP must be configured only if it provides measurable SEO or performance benefits for target landing pages given the chosen frontend stack.

If AMP is not recommended, the vendor must propose an alternative plan to achieve equivalent mobile performance and discoverability outcomes.

The preferred default assumption is:
- no AMP unless clearly justified,
- instead achieve performance via strong frontend architecture, rendering strategy, caching, media optimization, and page-weight control.

3.7 Analytics and Tracking (UX + Marketing)

3.7.1 Tracking Integration
The frontend must be compatible with:
- GA4,
- cookie consent,
- and a clean event model for funnel measurement.

The event model must support at minimum:
- page view by page type
- product list view
- product click
- shortlist / rail click
- bundle click
- add-to-cart or booking start
- checkout events
- purchase confirmation
- selected trust / hook interactions
- save / wishlist / compare behaviors if implemented

3.7.2 Tracking philosophy

Tracking must be:
- structured,
- stable,
- performance-safe,
- and aligned with the conversion funnel.

Tracking must not:
- excessively bloat the page,
- create unstable third-party script behavior,
- or undermine CWV performance.

3.8 Infrastructure and Delivery Requirements

3.8.1 CDN and Caching
The frontend must support CDN delivery and caching strategy appropriate for SSR / SSG / ISR and programmatic publishing.

Caching strategy must support:
- fast page delivery,
- predictable cache invalidation,
- controlled freshness,
- safe rollout of updated destination pages,
- and scalable publishing behavior.

3.8.2 Observability
Logs and APM must be available for monitoring:
- frontend performance,
- backend response times,
- rendering issues,
- API failures,
- and user-impacting errors.

3.8.3 CI/CD
A CI/CD pipeline must be implemented for:
- predictable release,
- rollback,
- environment control,
- and safe deployment of frontend changes.

Deployment quality must support:
- stable releases,
- emergency rollback,
- and compatibility with ongoing programmatic publishing.

B. Non-Functional Requirements (NFR)

B.1 Performance and Core Web Vitals

B.1.1 Goal
The service must be fast on mobile and meet strict Core Web Vitals targets.

Performance is part of the product experience, not a later optimization phase.

B.1.2 Core Web Vitals Targets (Mobile, P75)
Internal product target:
LCP (Largest Contentful Paint) must be below 1.5 seconds on:
- Home
- City
- Attraction / POI
- Product
- Packages / Bundles

CLS (Cumulative Layout Shift) must be below 0.1
INP (Interaction to Next Paint) must be below 200 ms

In addition, the platform must remain aligned with current Google public standards for good CWV performance on both mobile and desktop.

B.1.3 Backend Response Time Targets (No Cache)
P95 backend response time must be below 400 ms for HTML pages:
- City
- Product
- POI

P95 backend response time must be below 200 ms for JSON API endpoints:
- content payloads
- SEO payloads
- module/config payloads

P99 must not exceed 800 ms for HTML pages.

B.1.4 Frontend Page Weight Targets
Compressed JavaScript payload should not exceed 200–250 KB on Product and City pages.

Third-party scripts must be minimized and loaded in a performance-safe way.

The frontend should avoid script-heavy patterns that degrade:
- INP,
- hydration stability,
- or perceived responsiveness.

B.1.5 Image Requirements
Images must be delivered in WebP or AVIF where supported.
Lazy loading must be enabled by default.
Hero images must be preloaded.
Remaining images must be lazy-loaded.

Media must reserve layout space to avoid CLS.

B.1.6 Concurrency and Throughput
The system must support:
- at least 300 concurrent users
- at least 50 requests per second with stable response times
- horizontal scalability up to 5× these values via infrastructure scaling without code refactoring

B.1.7 Performance-safe UX behavior
The UX system must avoid:
- unstable layout shifts,
- re-render loops,
- heavy carousels without need,
- oversized media above the fold,
- and frontend patterns that look premium but degrade performance.

Premium must not come at the cost of responsiveness.

B.2 Accessibility and usability
The design system must support:
- strong readability,
- accessible contrast where required,
- keyboard navigation,
- screen-reader compatibility,
- large enough mobile tap targets,
- and predictable interaction states.

The site must feel premium and inclusive, not exclusive and difficult.

B.3 Reliability and compatibility
The frontend must behave consistently across:
- mobile browsers,
- desktop browsers,
- and common viewport ranges.

Critical booking and decision surfaces must not depend on fragile visual effects or device-specific hacks.

B.4 Rollback and resilience
The system must support:
- safe deployment,
- safe rollback,
- and graceful degradation when optional modules fail.

Critical flows such as:
- destination rendering,
- product rendering,

- availability initiation,
- and checkout entry

must continue to work even if optional modules are unavailable.

C. Expected Designer and UX Team Output

The design and UX team are expected to deliver:
- a modular design system,
- page templates for all core page types,
- clear decision-first layout hierarchy,
- premium visual standards,
- mobile-first flows,
- trust and proof placement logic,
- booking-flow UX,
- founder/brand trust module options,
- and experimentation-ready module variants.

They are expected not only to make the site “look good”, but to translate Rosotravel’s business model into a clear and convincing product experience.

The strongest outcome will be one where:
- users immediately feel premium quality,
- understand what the page is for,
- trust the shortlist,
- and move naturally from inspiration to booking.

END OF SCOPE 3